# ALGEBRA

Note Title 7/22/2026

TDN richiamo

1. $\boxed{\forall a, n, \quad n \mid \varphi(a^n - 1)}$.

$\cancel{a^n - 1 \equiv a - 1} \quad (n)$

Idee: $\cancel{\varphi(ab) = \varphi(a)\varphi(b)}$ $\quad \cancel{\varphi(a-1)}$

$\hookrightarrow$ $a^n - 1$ : Esp. mondu

$\operatorname{ord}_m(a) = \min k \text{ t.c. } a^k \equiv 1 \ (m)$

$(k, a^n - 1) \Rightarrow k^{\varphi(a^n-1)} \equiv 1 \quad (a^n - 1) \quad$ (PTF)

$n \overset{\text{hope}}{\mid} \operatorname{ord}_{(a^n-1)}(k) \mid \varphi(a^n - 1)$

Proviamo $\underline{k = a}$: la tesi è equiv. a $\boxed{n \mid \operatorname{ord}_{(a^n-1)}(a)}$

$\overset{a}{\text{Proviamo}}$

$\|$

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$\Downarrow$ $a^k \equiv 1\ (n)$ $\quad a^n - 1 \mid a^{n} - 1 \Rightarrow a^n \equiv 1\ (a^n - 1)$

$n = k \cdot \underbrace{\mathrm{ord}_{(a^n-1)}\, a}_{A}$, ma volevamo $n \mid \mathrm{ord}$

$a^n \equiv 1$

$n \mid A$, $A \mid n$ $\Rightarrow$ $\boxed{\mathrm{ord}_{(a^n-1)}\, a = n}$

$\forall k<n\ a^k \not\equiv 1\ (a^n - 1)$

$\times$ assurdo $a^k \equiv 1$

$a \equiv b\ (c) \rightarrow c \mid a - b$

$\boxed{a^n - 1 \mid a^k - 1}$ $\quad k < n$ assurdo

. FINE

## IMO 2026 / 1

2026 interi positivi $>1$ scritti alla lavagna,

$$(m,n) \longmapsto \left( \cancel{\gcd}^{\,MCD}(m,n),\ \frac{\cancel{lcm}^{\,mcm}(a,b)}{\cancel{\gcd}_{\,MCD}(a,b)} \right),$$

$$mcm = \frac{ab}{MCD} \quad \hookrightarrow \frac{ab}{(MCD)^2}$$

(a) dopo finite e arbitrarie mosse, $\exists!\, n > 1$.

(b) No

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$\boxed{\text{Sol}}$ $(a,b) = 1 \implies (a,b) \longrightarrow (1, ab)$

**Oss.** non tutti 1 alla fine, perché $(1,1) \longleftarrow (1,1)$

**Lemma** se abbiamo una situazione $\underbrace{1,1,1, \ldots, 1}_{2026-k}, \underbrace{k_1, \ldots, k_k}_{k}$ si ha la tesi. $[(k_i, k_j) = 1 \ \forall i,j]$

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$(m, n) \rightsquigarrow \text{mcm}(m, n)$ $\qquad a_1, \ldots, a_{2026}$

$\boxed{\prod a_i \text{ decresce debolmente}}$

$\hookrightarrow (a_i, a_j) = 1 \quad = \quad \Rightarrow$ non all'infinito

$(a_i, a_j) \neq 1 \quad <$

$\hookrightarrow \prod a_i$ decresce. A UN CERTO PUNTO SI FERMA

$\downarrow (1, 1, \ldots, 1)$ ASSURDO se all'infinito

nel caso

$\underbrace{1\ 1\ 1 \ldots}_{1} \quad \underbrace{u_1 \ldots u_p}_{\neq 1}$

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# VIETE

↳ POLINOMI DI QUALSIASI GRADO n

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x^1 + a_0$$

VIETE:

$$\sum_{i=1}^{n} \lambda_i = -\frac{a_{n-1}}{a_n}$$

$$\sum \lambda_i \lambda_j = +\frac{a_{n-2}}{a_n}$$

$$\sum \lambda_i \lambda_j \lambda_k = -\frac{a_{n-3}}{a_n}$$

$$\prod \lambda_i = (-1)^n \frac{a_0}{a_n}$$

DIm → GRADO 2 $n=2$

$$p(x) = a_2x^2 + a_1x + a_0$$

$\lambda_i$ = SOLUZIONE DI $P(x)$

$$p(x) = (x-\lambda_1)(x-\lambda_2)$$

$$= x^2 - \lambda_2 x + \lambda_1\lambda_2 - \lambda_1 x$$

$$= x^2 - x\underbrace{(\lambda_1+\lambda_2)}_{a_1} + \underbrace{\lambda_1\lambda_2}_{a_0}$$

$$\sum \lambda_i = -\frac{a_1}{a_2} = -\frac{a_1}{1} = -a_1 \qquad \prod \lambda_i = (-1)^n \frac{a_0}{a_n} = (-1)^2 \frac{a_0}{1} = a_0$$

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$$P(x) = (x-\lambda_1)(x-\lambda_2)\dots(x-\lambda_n)$$

SE IL POLINOMIO È MONICO

↳ IL COEF. DI GRADO MAX È 1

$$P(x) = 1x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1 x^1 + a_0$$

VIETE × POLINOMIO MONICO

$$\sum x_i = -\frac{a_{n-1}}{a_n} = -a_{n-1}$$

$$\sum \lambda_i \lambda_j = \frac{a_{n-2}}{a_n} = a_{n-2}$$

$$\prod \lambda_i = (-1)^n \frac{a_0}{a_n} = (-1)^n \cdot a_0$$

ES.

$$p(x) = 3x^4 + 5x^2 - 2x + 1$$

$$\sum \lambda_i = -\frac{a_{n-1}}{a_n} = -\frac{a_3}{a_4} = \frac{0}{3} = 0$$

$$\sum \lambda_i \lambda_j = \frac{a_{n-2}}{a_n} = \frac{5}{3}$$

$$\prod \lambda_i = (-1)^4 \frac{a_0}{a_n} = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

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$$p(x) = 3x^3 + 5x^2 - 2x + 1$$

$\lambda_1, \lambda_2, \lambda_3 \rightarrow$ SOLUZIONI $\qquad \lambda_1^2 + \lambda_2^2 + \lambda_3^2 = ?$

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$$\lambda_1^2+\lambda_2^2+\lambda_3^2=\underbrace{(\lambda_1+\lambda_2+\lambda_3)}_{-\frac{a_{n-1}}{a_n}}{}^2-2\underbrace{(\lambda_1\lambda_2+\lambda_2\lambda_3+\lambda_1\lambda_3)}_{\frac{a_{n-2}}{a_n}}$$

$$\lambda_1^2+\lambda_2^2+\lambda_3^2=-\frac{a_{n-1}}{a_n}-2\left(\frac{a_{n-2}}{a_n}\right)$$

$$=-\frac{5}{3}-2\left(\frac{-2}{3}\right)=-\frac{1}{3}$$

$$\frac{1}{\lambda_1^2}+\frac{1}{\lambda_2^2}+\frac{1}{\lambda_3^2}=\frac{\lambda_1^2\lambda_2^2+\lambda_1^2\lambda_3^2+\lambda_2^2\lambda_3^2}{\underbrace{\lambda_1^2\lambda_2^2\lambda_3^2}_{(P)^2\Rightarrow(-1)^n\frac{a_0}{a_n}}}=$$

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$$\overset{A}{(\lambda_1\lambda_2)^2+(\lambda_1\lambda_3)^2+(\lambda_2\lambda_3)^2} =$$

$$\underbrace{(\lambda_1\lambda_2+\lambda_1\lambda_3+\lambda_2\lambda_3)^2} - 2\underbrace{(\lambda_1^2\lambda_2\lambda_3+\lambda_1\lambda_2^2\lambda_3+\lambda_1\lambda_2\lambda_3^2)}$$

$$A - 2(\lambda_1\lambda_1\lambda_2\lambda_3+\lambda_1\lambda_2\lambda_2\lambda_3+\lambda_1\lambda_2\lambda_3\lambda_3)$$

$$A - 2\underbrace{(\lambda_1\lambda_2\lambda_3)}_{P}\underbrace{(\lambda_1+\lambda_2+\lambda_3)}_{S}$$

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Fatti sui polinomi $\mathbb{Z}[x]$

Lemma di Gauss

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$P(x), Q(x) \in \mathbb{Z}[x]$ primitivi $\Rightarrow P(x)Q(x)$ primitivo

Sketch dim

FTSOC $\exists p \mid a_i b_i \ \forall i$. Considero min $a_r, b_s \not\equiv 0 \ (p)$

$c_{r+s} = \sum a_i b_{r+s-i} = (\quad)(\quad) + \underbrace{\quad}_{\not\equiv 0 \ (p)} \not\equiv 0 \ (p)$

$\Rightarrow$ assurdo.

$(x^2+x+1)(x+2)$

$P(x), Q(x)$ irriducibile in $\mathbb{Z} \Rightarrow$ lo sono anche in $\mathbb{Q}$.

$\boxed{x^2-4} = 0$ $\quad x = 1$

# Lemma di Eisenstein

~~Supp~~ $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$. $\quad \mathbb{Z}[x]$

Supp. $\exists$ $p$ primo t.c. $p \mid a_{n-1}, a_{n-2}, \dots, a_0$, $\begin{cases} p \nmid a_n \\ p^2 \nmid a_0 \end{cases}$.

- Allora $p(x)$ è irriducibile.

es: $3x^4 + 5x^3 + 55x^2 + 60x + 15$

es$_2$: $2x^4 + 17x^3 + 102x^2 + 85x + 272$

Mason - Stothers (S21 M-A1) vedi [RMM 2019/2]

$A(x)$, $B(x)$, $C(x)$ coprimi tra loro.

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$\max(\deg(A), \deg(B), \deg(C)) < \#\text{radici}(A(x)B(x)C(x))$

$\leq \sum_{\text{cyc}} \left(\#\text{radici}(A(x))\right).$

$\mathbb{R}, \mathbb{R}[x]$

2

3 $\mapsto$ $2+3+4$

4

esempi: $P(x) = x^3 + 12x - 6x^2 - 8 = (x-2)^3$

$Q(x) = x^2 + 5$

$R(x) = x + 1$

Quante radici ha $PQR$?

$(x-2)^3(x-1)\underbrace{R(x)}_{\text{irriducibile}}$

$\#\text{radici}(PQR) \geq 3 \Rightarrow \boxed{2, 4}$

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AM - GM

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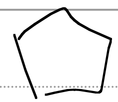
Media Aritm. - Media Geom.

$$\frac{x_1+x_2+x_3 \dots +x_n}{n} \rightarrow \sqrt[n]{x_1 x_2 x_3 \dots x_n}$$

$$\boxed{\frac{a+b}{2} \geq \sqrt[2]{ab}} \rightarrow AM \geq GM$$

$$\rightarrow a+b \geq 2\sqrt[2]{ab}$$

$$a+b-2\sqrt{ab} \geq 0$$

$(\sqrt{a} - \sqrt{b})^2 \geq 0$ 

$\downarrow$

$a + b - 2\sqrt{ab}$

$\downarrow$

Ex. $a = 1 \quad b = x$

$$\frac{1+x}{2} \geq \sqrt{x \cdot 1}$$

Es. $(a+b)(b+c)(c+a) \geq 8abc$

AM-GM

$a + b \geq 2\sqrt{ab}$

$\frac{a+b}{2} \geq \sqrt{ab} \quad a+b \geq \sqrt{ab}$

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$b+c \geq 2\sqrt{bc}$

$c+a \geq 2\sqrt{ca}$

$\hookrightarrow (a+b)(b+c)(c+a) \geq 2\sqrt{ab} \cdot 2\sqrt{bc} \cdot 2\sqrt{ca}$

$\geq 8abc$

NOTAZIONE DISUGUAGLIANZE

LHS - RHS

$\leq$

$=$

$\geq$

$\hookrightarrow AM = \dfrac{x_1 + x_2 + \dots + x_n}{n} \qquad GM = \sqrt[n]{x_1 \cdots x_n}$

es.

$$1 + x^2 + x^4 + x^6 \cdots + x^{202} \geq 102\, x^{101}$$

$$1 + x^2 + x^4 + x^6 + \cdots + x^{202} \geq 102 \sqrt[102]{1 \cdot x^2 \cdot x^4 \cdot x^6 \cdots x^{202}}$$

$$\sqrt[102]{x^2 \cdot x^4 \cdot x^6 \cdots x^{202}}$$

$$\left(x^2 \cdot x^4 \cdot x^6 \cdots x^{202}\right)^{\frac{1}{102}}$$

$$(x)^{\frac{1}{102} \cdot (2 + 4 + 6 + \cdots + 202)}$$

$\mathbb{R}^+$

$$x^{\frac{1}{102}\cdot 2} + x^{\overline{102}\cdot 4}$$

$$(x^2)^{102} + (x^4)^{102}$$

$$(x)^{\frac{1}{102}\cdot\underbrace{(2+4+6+\dots+202)}}$$

$$(x)^{\frac{1}{102}\cdot 2\underbrace{(1+2+3+\dots+101)}_{\frac{n(n+1)}{2}=}}$$

$$(x)^{\frac{1}{\cancel{102}}\cdot\cancel{2}\cdot\frac{101\cdot\cancel{102}}{\cancel{2}}} = x^{101}$$

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$\geq$

DISUGUAGLIANZE PESATE

det il min. $x^4 + y^2 + z$ $\quad x, y, z \in \mathbb{R}^+$ $\quad xyz = \frac{1}{2}$

$x^4 + y^2 + z \geq 3\sqrt[3]{x^4 y^2 z}$

$z = \frac{z}{4} + \frac{z}{4} + \frac{z}{4} + \frac{z}{4}$ $\quad x^4 \quad z^1$

$$y^2 = \frac{y^2}{2} + \frac{y^2}{2}$$

$$x^4 + y^2 + z = \underbrace{x^4 + \frac{y^2}{2} + \frac{y^2}{2} + \frac{z}{4} + \frac{z}{4} + \frac{z}{4} + \frac{z}{4}}$$

AM-GM

$$\text{LHS} \geq 7\sqrt[7]{x^4 \cdot \frac{y^2}{2} \cdot \frac{y^2}{2} \cdot \frac{z}{4} \cdot \frac{z}{4} \cdot \frac{z}{4} \cdot \frac{z}{4}}$$

$$\text{LHS} \geq 7\sqrt[7]{\frac{x^4 y^4 z^4}{2^{10}}} \qquad xyz = \frac{1}{2}$$

$$7\sqrt[7]{\frac{(xyz)^4}{2^{10}}} = 7\sqrt[7]{2^{-14}} = 7 \cdot \frac{1}{4} = \frac{7}{4}$$

# Media $p$-esima

$$\sqrt[p]{\frac{a_1^p + a_2^p + \ldots + a_n^p}{n}}$$

$\underline{p = 1} \Rightarrow AM$

$\underline{p = 0} \Rightarrow GM$

$\underline{p=2} \Rightarrow \sqrt{\frac{\sum a_i^2}{n}}$ QM

$\underline{p=-1} \Rightarrow \left(\frac{\frac{1}{a_1} + \frac{1}{a_2} + \ldots + \frac{1}{a_n}}{n}\right)^{-1} = \frac{n}{\frac{1}{a_1} + \ldots + \frac{1}{a_n}}$ HM

Vale: $\underset{2}{QM} \geq \underset{1}{AM} \geq \underset{0}{GM} \geq \underset{-1}{HM}$.

In generale $M_p \geq M_q$ se $p \geq q$.

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## Cauchy - Schwartz

2 n-uple $\mathbb{R}$ $a_1, \ldots, a_n$ . Allora
$b_1, \ldots, b_n$

$$\left(\sum a_i^2\right)\left(\sum b_i^2\right) \geq \left(\sum a_i b_i\right)^2 .$$

$n=2$ : $(a_1^2 + a_2^2)(b_1^2 + b_2^2) \geq (a_1 b_1 + a_2 b_2)^2$.

TBC

## TdA Senior 2024

↳ det. min $u$ t.c. $a^2 + \frac{2}{b} + 3c \leq \sqrt{a^2+u}\,\sqrt{a^2+\frac{1}{b^2}+c^2}$

⇓

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$\boxed{\text{Sol}}$ $(a^2+4)\left(a^2+\frac{1}{b^2}+c^2\right)$. Vogliamo C-S ...

ci hanno mandato via dall'aula.
VEDASI FOTO LAVAGNA PER CONCLUSIONE

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